Shaeekul Ameen

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PROFESSIONAL EXPERIENCE

Graduate Research Assistant
Gainesville, Fl, USA
Pico scale Engineering Research Lab (PERL) *August 2025 –Present*

My research focuses on advanced electron microscopy techniques, specifically ptychography and atomic electron tomography (AET), for high-resolution structural characterization of nanoscale materials. My work involves phase retrieval from diffraction data, 3D atomic reconstruction from tilt-series projections, and quantitative analysis of nanoscale defects and lattice distortions.

Military Institute of Science & Technology
Dhaka, Bangladesh
Lecturer – Full Time, Dept. of Electrical, Electronic & Communication Engineering *August 2023 –June 2025*

Course taught:

- Fundamentals of Electrical Circuit Analysis (Sessional)
- Electrical and Electronics Technology (Sessional).
- Electronic Circuits and Devices (Sessional).
- Communication Theory I (Sessional).
- VLSI I (Sessional & Theory). - VLSI Design (CMOS logic design, layout principles, scaling), Digital Integrated Circuits (timing, power, noise analysis), Digital Logic Design (combinational & sequential circuits).
- Control System I (Sessional).
- Processing and Fabrication Technology (Theory).

Professional activity:

- Member, OBE Quality Assurance and BAETE Accreditation Committee
- Member, Technical & Organizing Committee, 6th International Conference on Electrical, Electronics, Information and Communication Technology (ICEEICT)
- Assistant officer in charge: VLSI Laboratory
- Assistant Course Coordinator (ACC- EECE 19)

Southeast University
Dhaka, Bangladesh
Lecturer – Part Time, Dept. of Electrical &, Electronic Engineering *January 2024 –June 2025*

Course taught:

- Processing and Fabrication Technology (Theory).
- VLSI - I (Theory).
- Digital Signal Processing (Sessional).
- Communication Engineering (Sessional).

Manarat International University
Dhaka, Bangladesh
Lecturer – Part Time, Dept. of Electrical & Electronic Engineering *January 2024 –May 2024*

Course taught:

- Power system - II (Theory).
- Electronics - I (Sessional).

EDUCATION

University of Florida
Florida, USA
M.Sc. in Electrical & Computer Engineering *August 2025 – Present*
CGPA: 3.91/4.00 (Ongoing)

Ahsanullah University of Science & Technology
Dhaka, Bangladesh
B.Sc. in Electrical & Electronic Engineering *April 2018 – December 2022*
CGPA: 3.861/4.00
Undergrad Thesis: Quantum Simulation Study of Carbon Nanotube Field Effect Transistor with High 'k' Di-electrics

ACADEMIC HONORS

- Dean's List of Honor in all four academic years *2022*

PUBLICATIONS

- "*Parametric Dependency of Charge Transport in a Carbon Nanotube-Based Field Effect Transistor: A Numerical Simulation*". 2023 IEEE 8th International Conference for Convergence in Technology (I2CT), Lonavla, India, 2023, pp. 1-6
- "*Parametric Simulation Study of Control Coefficients on the On-Off Current Ratio of Single-Walled Carbon Nanotube Field Effect Transistor*". European Journal of Electrical Engineering and Computer Science. 7, 5 (Oct. 2023)
- "*Performance Analysis of Thickness Dependency of Control Coefficients of CNTFET*". Conference: 2024 6th International Conference on Electrical Engineering and Information & Communication Technology.
- "*Numerical Analysis of Diameter Dependency of Control Coefficient of Carbon Nanotube Field Effect Transistor*". Majlesi Journal of Electrical Engineering 18(2):1-7.

ACADEMIC PROJECTS

- *Atomic-Scale Characterization of Semiconductor Nanostructures Using Electron Ptychography and Atomic Electron Tomography*
  Developed electron ptychography reconstruction and analysis workflows using PtyRAD, abTEM, Atomap, and HyperSpy. Performed atomic column detection, Gaussian fitting, lattice spacing measurements, and strain mapping for quantitative characterization of Si/SiGe semiconductor interfaces. Optimized reconstruction parameters to improve atomic-resolution imaging and analysis accuracy.
- *CMOS Full Adder Design, Layout, and Performance Analysis*
  Designed and analyzed 4 CMOS full adder cells using transistor-level techniques; implemented XOR/multiplexer-based logic, performed schematic simulation, layout with DRC/LVS verification, and evaluated power–delay trade-offs.
- *Underground Fault detection device*
  Using Arduino Uno faults in power line system was detected and analyzed at several distances.
- *Noise Reduction using MATLAB*
  Using MATLAB to record voice, filter the recording through mathematical analysis and reduce the background noise.
- *Wireless DC motor control using generic remote*
  Controlling a DC motor with a remote control, a 555 timer, or power electronics devices is useful for day-to-day device control.
- *Two-way traffic system using Arduino Uno*
  Building a traffic control system that helps control the traffic in congested areas, keeping in mind that the whole system should be cheap and easy to maintain.

RESEARCH INTEREST

- Electron Ptychography, 4D-STEM, Atomic Electron Tomography, Computational Microscopy, Semiconductor Device Characterization, Semiconductor Device Physics and Nanoelectronics, Carbon nanotubes, VLSI, Nano fabrication, Nanoelectronics, Optoelectronics, Bio Electronics, CNTFET

TECHNICAL SKILLS

- Programming Languages: Python (abTEM, PtyRAD, Atomap, HyperSpy, NumPy, SciPy, Pandas, Matplotlib), C, C++, MATLAB, Verilog, HTML
- Electron Microscopy & Computational Imaging: Electron Ptychography, 4D-STEM, Atomic Electron Tomography (AET), HAADF-STEM, Multislice Reconstruction, Phase Retrieval, Atomic Column Detection, Gaussian Peak Fitting, Strain Mapping, Lattice Analysis
- Tools: Google Workspace, Git, Visual Studio, Codeblocks, MATLAB, Arduino
- Simulators: Orcad Pspice, Quartus, Proteus, Cadence, AutoCad, TinkerCad.
- Office tools: Microsoft Word, PowerPoint, Excel

PROFESSIONAL TRAINING

- Basic Safety Training, Samsung C&T Safety Academy *December 2022*

SYNERGISTICS ACTIVITIES

- Member, AUST EEE Society (Organizer) *2022*
- Participant in Walton laptop represents Mindspark'22 *2022*
- Participant in Bangladesh Model United Nations *2017*
- Winner in Project Showcase, Notre Dame Science Festival *2017*
- Project Finalist, Green Inspiration International Summit *2016*
- Cub Scout, Shapla-Cub-Award winner *2009*

REFERENCE

1. **Dr. Navid Asadi**
   Associate Professor
   Department of Electrical & Computer Engineering
   University of Florida, Florida - USA
   Email: nasadi@ece.ufl.edu

2. **Dr. Dennis Kim**
   Assistant Professor
   Department of Electrical & Computer Engineering
   University of Florida, Florida - USA
   Email: ds.kim@ufl.edu

3. **Md. Faysal Nayan**
   Assistant Professor
   Department of Electrical & Electronic Engineering
   Ahsanullah University of Science & Technology,
   Email: faysal.nayan.eee@aust.edu